Hybrid versus intrinsic Majorana nanowires, a self-consistent Hartree-Fock-Bogoliubov analysis

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Not yet worthy nor ready for the abstract.

I. INTRODUCTION

Not yet worthy nor ready for the introduction.

II. SELF-CONSISTENT HARTREE-FOCK-BOGOLIUBOV THEORY

Consider a generic tight-binding quartic Hamiltonian,

$$\hat{H} = \sum_{\alpha\beta} h_{\alpha\beta} c^\dagger_\alpha c_\beta + \frac{1}{2}\sum_{\alpha\beta\gamma\delta} V^{\alpha\beta}_{\gamma\delta} c^\dagger_\alpha c^\dagger_\beta c_\gamma c_\delta, \tag{1}$$

with $c^\dagger_\alpha(c_\alpha)$ fermionic creation (annihilation) operators of site and spin degree of freedom $\alpha = (i, \sigma_i)$, and $V^{\alpha\beta}_{\gamma\delta}$ a spinless electrostatic potential tensor of symmetries $V^{\alpha\beta}_{\gamma\delta} = V^{\beta\alpha}_{\delta\gamma}$ and $(V^{\alpha\beta}_{\gamma\delta})^* = V^{\gamma\delta}_{\alpha\beta}$ following directly from $V(\mathbf{r}-\mathbf{r}') = V(\mathbf{r}'-\mathbf{r})$. We fabricate a doubling of the degrees of freedom by introducing the Nambu-spinor $\check{c}_a = (c_a \; c^\dagger_a)$, constraining a particle-hole symmetry. In this Nambu representation the system's reduced density matrix (rDM) corresponds to the tensor product $\rho_{ab} = \langle \check{c}^\dagger_b \otimes \check{c}_a \rangle$, reading explicitly as

$$\rho_{ab} = \begin{pmatrix} \langle c^\dagger_b c_a\rangle & \langle c_b c_a\rangle \\ \langle c^\dagger_b c^\dagger_a\rangle & \langle c_b c^\dagger_a\rangle \end{pmatrix} = \begin{pmatrix} \rho^{\rm ee}_{ab} & \rho^{\rm eh}_{ab} \\ \rho^{\rm he}_{ab} & \rho^{\rm hh}_{ab} \end{pmatrix}, \tag{2}$$

with it's terms relating to each other directly from the fermionic anti-commutator properties.

Performing a mean-field Hartree-Fock decoupling as follows $\langle c^\dagger_\alpha c^\dagger_\beta c_\gamma c_\delta\rangle \approx \rho^{\rm he}_{\beta\alpha}\rho^{\rm eh}_{\delta\gamma} - \rho^{\rm ee}_{\gamma\alpha}\rho^{\rm ee}_{\delta\beta} + \rho^{\rm ee}_{\delta\alpha}\rho^{\rm ee}_{\gamma\beta}$, the Nambu rDM equation of motion can be shown to read simply as $i\hbar\dot{\rho}_{ab} = \left[H^{\rm BdG}_{ab}, \rho_{ab}\right]$ with

$$H^{\rm BdG}_{ab} = \begin{pmatrix} h^{\rm HF}_{ab} & \Sigma^\Delta_{ab} \\ (\Sigma^\Delta_{ba})^* & -h^{\rm HF}_{ba} \end{pmatrix} \tag{3}$$

the effective Bogoliubov-de Gennes (BdG) Hamiltonian, $h_{\rm HF}(\rho) = h + \mathbf{\Sigma}_H(\rho) + \mathbf{\Sigma}_F(\rho)$ the Hartree-Fock Hamiltonian, having defined the Hartree, Fock and pairing self-energies as:

$$\begin{matrix} \Sigma^H_{ab} = \sum_{\gamma\delta} V^{\delta a}_{b\gamma}\rho^{\rm ee}_{\gamma\delta} \\ \Sigma^F_{ab} = -\sum_{\gamma\delta} V^{a\delta}_{b\gamma}\rho^{\rm ee}_{\gamma\delta} \end{matrix} \quad \text{and} \quad \Delta_{ab} = \sum_{\gamma\delta} V^{ab}_{\delta\gamma}\rho^{\rm eh}_{\gamma\delta} \tag{4}$$

Note the pairing symmetry $\Sigma^\Delta_{ab} = -\Sigma^\Delta_{ba}$.

We build upon these results by further realizing the mean-field self-energy as a spin-1/2 Hubbard term in the Wannier limit, i.e considering the approximation of ultra-localized orbitals $V^{ij}_{kl} \approx V^{ij}\delta_{il}\delta_{jk}$ with a spinless potential $V^{i\sigma_i j\sigma_j} = V^{ij}$, followed by the exclusion of all offsite term contribution through $V^{ij} = U\delta_{ij}$ with $U < 0$ an attractive Hubbard potential. The Nambu-spinor follows as $\check{\boldsymbol{c}}_i = (c_{i\uparrow} \; c_{i\downarrow} \; c^\dagger_{i\uparrow} \; c^\dagger_{i\downarrow})$. Defining a new BdG symmetric rDM object as

$$\tilde{\boldsymbol{\rho}}_{ij} = \begin{pmatrix} \boldsymbol{\rho}^{\rm ee}_{ij} & -\boldsymbol{\rho}^{\rm eh}_{ij} \\ -\left(\boldsymbol{\rho}^{\rm eh}_{ji}\right)^* & -\boldsymbol{\rho}^{\rm ee}_{ji} \end{pmatrix}, \tag{5}$$

allows one to write the total onsite self-energy of the system in the compact form

$$\mathbf{\Sigma}_{ij} = \frac{1}{2}U\left[{\rm Tr}\left(\left[\tau_z\otimes\sigma_0\right]\tilde{\boldsymbol{\rho}}_{ii}\right)\left[\tau_z\otimes\sigma_0\right] - 2\tilde{\boldsymbol{\rho}}_{ij}\right]\delta_{ij}. \tag{6}$$

with both σ and τ being Pauli matrices operating in their respectively spin and electron-hole subspaces and the Kronecker deltas δ_{ij} denoting on-site terms and $\delta^\pm_{ij\pm1}$ the 1st neighbors hoppings (front and back). Note that the underlying spin structure is being concealed within the bold notation.

III. NANOWIRE MODELS

First and foremost, we showcase Oreg-Lutchyn minimal Majorana model [?] as our model's groundwork. Oreg-Lutchyn minimal model consists of a finite one-dimensional nanowire in the presence of an external Zeeman field $V_{\rm Z}$ and spin-orbit coupling (SOC) effect, being described by the Hamiltonian

$$\hat{\boldsymbol{H}}_0 = \sum_{\sigma_i\sigma_j} c^\dagger_{\sigma_i}\left(-\eta\partial^2_x - \mu + \alpha_{\rm R} i\sigma_y\partial_x + V_{\rm Z}\sigma_z\right)c_{\sigma_j}, \tag{7}$$

with $\eta = \hbar^2/2m^*$, m^* the electron's effective mass, and μ the chemical potential. The SOC effect due an externally applied transverse magnetic field of Rashba strength $\alpha_{\rm R}$ and the Zeeman splitting $V_{\rm Z} = g_J\mu_B B_x/2$ due to a in-plane magnetic field B_x, with g_J the Landé $g$yromagnetic factor of the semiconductor and μ_B Bohr's magneton. Furthermore, the minimal model then considers a s-type superconductive (SC) proximity effect of uniform pairing Δ_0 reading as $(\Delta_0 c^\dagger_\uparrow c^\dagger_\downarrow + h.c)/2$.

The major step up to the minimal model is that we do not consider an homogeneous superconductive proximity effect term but instead self-consistently calculate for the induced pairing (but also magnetization and doping).

As derived in the previous section, this is realized in a Hartree-Fock-Bogoliubov mean-field theory. Within the Nambu tight-binding scheme, this correction amounts to the substitution $\Delta_0 \left[\tau_y \otimes \sigma_y\right] \delta_{ij} \to \mathbf{\Sigma}_{ij}$, with $\mathbf{\Sigma}_{ij}$ made explicit in Eq.(6). The full Hamiltonian matrix reads explicitly as

$$\boldsymbol{H} = \left(f\delta_{ij} - t\delta_{ij\pm1}\right)\left[\tau_z \otimes \sigma_0\right] + V_{\mathrm{Z}}\left[\tau_z \otimes \sigma_z\right]\delta_{ij} + \frac{\alpha_{\mathrm{R}}}{2a_0}\delta^{\pm}_{ij\pm1}\left[\tau_z \otimes i\sigma_y\right] + \mathbf{\Sigma}_{ij} \tag{8}$$

with $f = 2t - \mu$ the band filling ($\mu = 0$ and $\mu = 2t$ corresponding to half and zero filling respectively), $t = \eta/a_0^2$ the hopping parameter and a_0 the real-space lattice spacing.

Schematic models of the hybrid and intrinsic self-consistent Majorana nanowires are shown in Fig.(1).

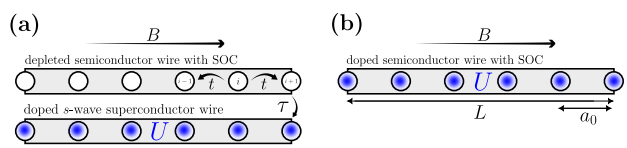

Figure 1. **Self-consistent finite nanowire models. (a) Hybrid model.** A depleted semiconducting (SM) nanowire in a doped s-wave superconducting (SC) nanowire substrate. Both nanowires are subject to an externally applied magnetic field B but only the SM exhibits spin-orbit coupling. Crucially, only the SC nanowire contains the self-consistent Hubbard term U_{SC} (therewith $U_{\mathrm{SM}} = 0$), thus serving as the sole source of pairing within the system. The proximity effect is modeled by a hopping with intensity $\tau < t$. Moreover, note that considering $\mu_{\mathrm{SC}} \gg \mu_{\mathrm{SM}}$ is crucial for an effective proximity effect because we do not account for disorder and the parallel momentum must be conserved across the interface. **(b) Intrinsic model.** Just a doped semiconductor wire (no SC subtract) with strong SOC subjected to an externally applied magnetic field B. The superconductivity arrives intrinsically due to the self-consistent Hubbard term U.

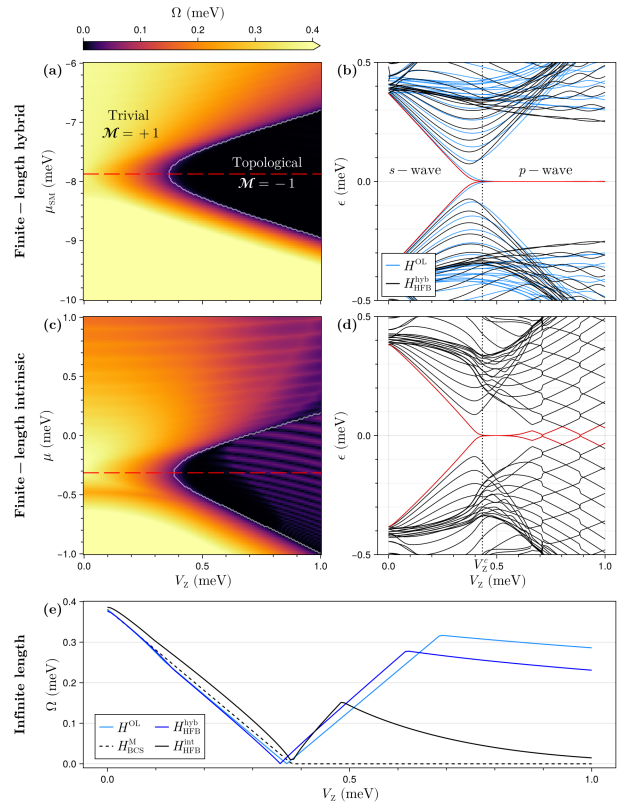

Figure 2. **Self-consistent nanowire models results.** The white boundary line denotes the abrupt change in the Majorana number $\mathcal{M}$ marking the trivial-topological phase transition. Common to all models, we have set the lattice spacing to $a = 10$nm, the finite nanowire length to $L = 2000$nm, the effective mass to $m^* = 0.023m_e$ and the SOC Rashba strength to $\alpha = 40$meV. For the hybrid case the SC Hubbard strength is $U_{\mathrm{SC}} = -30$meV with the SC chemical potential $\mu_{\mathrm{SC}} = 5$meV and SC-SM hopping coupling $\tau = 0.8t$. For the intrinsic case the Hubbard strength is $U = -8$meV.

IV. RESULTS

Somewhat worthy, but better if you write it. Here are some not so obvious small talking points such that you don't forget them:

1. Shallow mode μ shift.

2. Self-consistent routine.

   - Full spectrum diagonalization for finite case
   - Schur's factorization for infinite case
   - Anderson acceleration method

3. Analytical BCS gap equation:

   Furthermore, we compare the μ_0 linecuts to the analytical BCS gap equation, generalized to account for Zeeman but not SOC, derived in appendix:

$$\Delta_0 = 2\epsilon_0 \exp\left(-\sqrt{\frac{\pi^2}{2}\frac{\hbar^2}{m_e e}}\sqrt{\frac{\mu}{m_0}}\frac{1}{|U|a_0}\right), \tag{9}$$

   It is important to note that Eq.(9) assumes $\mu \gg \Delta$. See that the actual spectral gap Ω (show in dashed black) does not always correspond to the pairing potential Δ_0. Generally it reads $\Omega = \max(0, |\Delta_0| - |V_{\mathrm{Z}}|)$ since for $|V_{\mathrm{Z}}| > \Delta_0$ the gap has to close although the pairing remains finite. Moreover, see that one always needs $\mu > 0$ or else the pairing becomes non-real. In the particular case $V_{\mathrm{Z}} = 0$ with $\mu < 0$ the gap must correspond instead to $\Omega = -\mu$.

4. Direct analysis from the figure.

   - Topological invariant, OL hyperboloid
   - Self-induced dopping shifts μ_0^{hyb} and $\mu_0^{\mathbf{int}}$

- Mirror-symmetric holeonic phase-transition hyperboloid
- Insulator region in the phase diagram
- No Majorana oscillation because large L

5. Refer to the multi-mode intrinsic results.

V. CONCLUSIONS

Not yet worthy nor ready for the conclusions.

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APPENDIX

Appendix A: Analytical gap equation

The main objective of this section is to derive the typical textbook BCS gap equation for an infinite system without Zeeman or SOC [??] from our Hartree-Fock-Bogoliubov approach. We generalize the result for a Zeeman field, accounting not only for the self-consistent pairing but also the self-consistently induced magnetization and doping. We will not, however, account for SOC effects.

Foremost, let us reparametrize the self-energy of Eq.(6) in terms of the self-consistently induced magnetization $\delta V_{\rm Z}$, doping $\delta\mu$ and pairing Δ (assumed to be real) as $\mathbf{\Sigma}_{ii} = (\delta V_{\rm Z}\left[\tau_z\otimes\sigma_z\right] - \delta\mu\left[\tau_z\otimes\sigma_0\right]) - \Delta\left[\tau_y\otimes\sigma_y\right]$. Comparing this rewritten form with the original one, we can identifying the off-diagonal block $\Sigma^{\rm he}_{ii}$ as being $-i\Delta\sigma_y = U\rho^{\rm eh}_{i\uparrow i\downarrow}$, noting that $\rho^{\rm eh}_{i\uparrow i\downarrow} = -\rho^{\rm eh}_{i\downarrow i\uparrow}$.

In another context, consider the Hamiltonian (no SOC) $H_0(k) = \sum_s c^\dagger_{ks}\left[(\epsilon_k-\mu)\delta_{ss'} + V_Z\sigma_z\right]c_{ks'}$ with s spin and k momentum. For the Hamiltonian at hand we have $\rho^{\rm eh}_{k\uparrow\downarrow} = \frac{1}{N}\sum_k\left[u^*_{k+}v_{k+}f(\tilde{E}_{k+}) + u^*_{k-}v_{k-}f(\tilde{E}_{k-})\right]$ where $\tilde{E}_{k\pm} = \tilde{V}_{\rm Z}\pm\tilde{E}_k$ with $\tilde{E}_k = \sqrt{\Delta^2+\tilde{\epsilon}^2_k}$ (no $\pm$ subindex) are its eigenvalues and $(u_{k\pm},\, v_{k\pm})$ the (normalized) eigenvectors of the $\uparrow\downarrow$ block; $\tilde{\epsilon}_k = \epsilon_k - \tilde{\mu}$, $\tilde{\mu} = \mu + \delta\mu$ and $\tilde{V}_{\rm Z} = V_{\rm Z} + \delta V_{\rm Z}$ already account for the U-induced shift and f is the Fermi-Dirac distribution. It is critical to realized that is in this generalization we have additional, implicit, self-consistent condition for both $\delta V_{\rm Z}$ and $\delta\mu$. These are obtained in a similar fashion to the pairing term but considering instead the diagonal elements of $\mathbf{\Sigma}_{ii}$. The self-consistent system of three equation for pairing Δ, induced magnetization $\delta V_{\rm Z}$ and doping $\delta\mu$ read as

$$\Delta = -U\sum_k\frac{\Delta}{4\tilde{E}_k}\left[\tanh\left(\frac{\tilde{V}_{\rm Z}+\tilde{E}_k}{2k_BT}\right) - \tanh\left(\frac{\tilde{V}_{\rm Z}-\tilde{E}_k}{2k_BT}\right)\right] \tag{10}$$

$$\delta V_{\rm Z} = -\frac{U}{2N}\sum_k\frac{\sinh\left(\frac{\tilde{V}_{\rm Z}}{k_BT}\right)}{\cosh\left(\frac{\tilde{V}_{\rm Z}}{k_BT}\right)+\cosh\left(\frac{\tilde{E}_k-\tilde{V}_{\rm Z}}{k_BT}\right)} \tag{11}$$

$$\delta\mu = \frac{U}{4} + \frac{U}{4N}\sum_k\frac{\tilde{\epsilon}_k}{2\tilde{E}_k}\left[\tanh\left(\frac{\tilde{V}_{\rm Z}+\tilde{E}_k}{2k_BT}\right) - \tanh\left(\frac{\tilde{V}_{\rm Z}-\tilde{E}_k}{2k_BT}\right)\right] \tag{12}$$

See that, if one ignores in Eq.(10) Zeeman all together, $V_{\rm Z}$ and $\delta V_{\rm Z}$, but also the self-consistently doping $\delta\mu$ it reduces to the traditional gap equation [??] as intended. Note that this has a nonzero solution only if the interaction is attractive i.e $U<0$. Although these analytical expressions can be obtained, the generalized analytical solution of the self-consistent system of equation is neither trivial nor worth the trouble for our purposes. We can, however, solve the conventional gap equation in the zero-temperature, small-gap limit, which offers a sufficiently accurate analytical comparison to our numerical results. Thereby, we approximate the dispersion to $\epsilon_k = \pm v_{\rm F}(k\mp k_{\rm F})$ with $v_{\rm F}$ and $k_{\rm F}$ the Fermi velocity and wavevector respectively, and mutate the $\frac{1}{N}\sum_k$ sum into an integral over ϵ_k, introducing also a fitted phenomenological cutoff ϵ_0. In the limit where $\Delta\ll\epsilon_0$ and for $U<0$ the gap equation simplifies to

$$\Delta = 2\epsilon_0\exp\left(-\sqrt{\frac{\pi^2}{2}\frac{\hbar^2}{m_e e}}\sqrt{\frac{\mu}{m_0}}\frac{1}{|U|a_0}\right). \tag{13}$$

Even though we neglected Zeeman this solution is still valid for finite $|V_{\rm Z}|\le\Delta$ since the tanh factors remain equal to 1 throughout the integration interval. Nevertheless, the generalization of Eq.(13) for $|V_{\rm Z}|>\Delta$ can be found from Eq.(10) within the same regime of approximation and expressed in terms of the previous $\Delta_0\equiv\Delta(V_{\rm Z}=0)$ solution as

$$\Delta = \frac{1}{2\epsilon_0}\sqrt{\Delta_0\left(2\left|V_{\rm Z}\right|-\Delta_0\right)}. \tag{14}$$

It is important to note that the actual spectral gap Ω does not always correspond to the pairing potential Δ_0. Generally it reads $\Omega = \max(0, |\Delta_0| - |V_{\rm Z}|)$ since for $|V_{\rm Z}|>\Delta_0$ the gap has to close although the pairing remains finite. Moreover, see that one always needs $\mu>0$ or else the pairing becomes non-real. In the particular case $V_{\rm Z}=0$ with $\mu<0$ the gap must correspond instead to $\Omega=-\mu$.